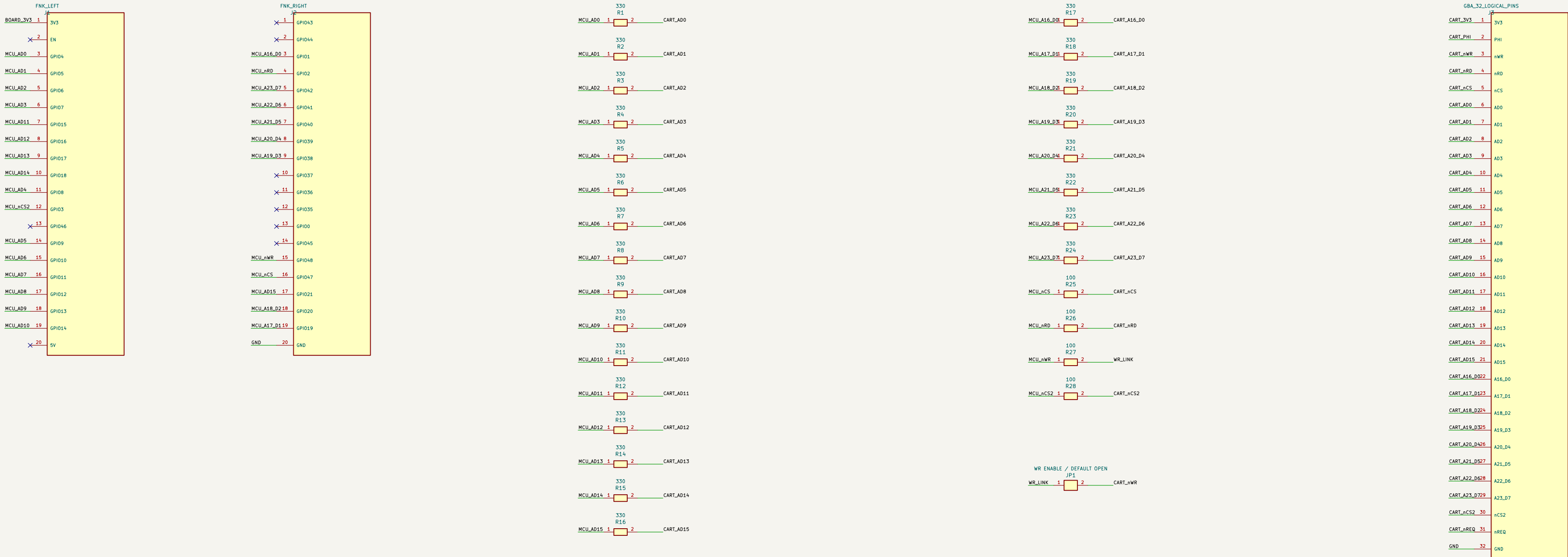


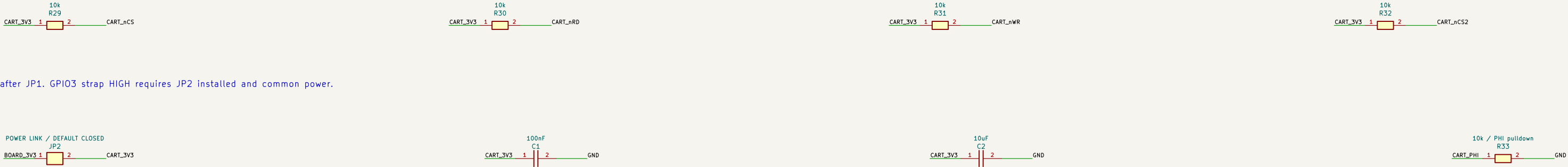
FNK0099 GBA DIRECT V1 / LOGICAL SCHEMATIC / NOT FOR FABRICATION

FNK0099_GBA_DIRECT_V1_20260927 | SMD removal permitted; targets NOT yet identified | GPIO35/36/37 ALWAYS NC

SOCKET VIEW: component side, antenna UP, USB DOWN: top pin = 1. All footprints require physical verification.



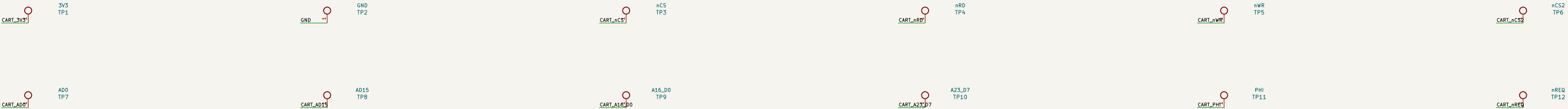
R1–R33, C1–C2: 0805 / JLCPCB Basic selected / electrical validation pending



CONTROL PULLUPS: cartridge side; R31 after JP1. GPIO3 strap HIGH requires JP2 installed and common power.

GBA ONLY / 3.3V | JP1/JP2 and cartridge: REMOVE USB FIRST | Native USB: DO NOT CONNECT – GBA BUS

Power source: FNK0099 J1–1. Regulator capability unverified. No external supply in parallel. No 5V cartridge connection.



/REQ: test point ONLY; no GPIO, no ground short. PHI: R33 to GND. USB–UART through on–board CH343 only.

ERC scope: passive carrier interconnect only. FNK0099 LED/USB/strap circuits, boot behavior and signal integrity need bench tests.